

**2015 HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–17

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is 0.005 233 59 written in scientific notation, correct to 4 significant figures?

- (A) 5.2336×10^{-2}
- (B) 5.234×10^{-2}
- (C) 5.2336×10^{-3}
- (D) 5.234×10^{-3}

2 What is the slope of the line with equation $2x - 4y + 3 = 0$?

- (A) -2
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) 2

3 The first three terms of an arithmetic series are 3, 7 and 11.

What is the 15th term of this series?

- (A) 59
- (B) 63
- (C) 465
- (D) 495

- 4 The probability that Mel's soccer team wins this weekend is $\frac{5}{7}$.

The probability that Mel's rugby league team wins this weekend is $\frac{2}{3}$.

What is the probability that neither team wins this weekend?

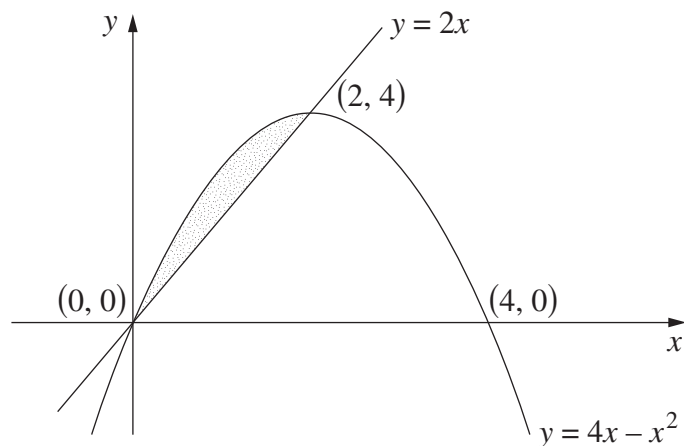
- (A) $\frac{2}{21}$
- (B) $\frac{10}{21}$
- (C) $\frac{13}{21}$
- (D) $\frac{19}{21}$
- 5 Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

- (A) $\frac{1}{4}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
- (B) $\frac{1}{4}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$
- (C) $\frac{1}{2}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
- (D) $\frac{1}{2}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

6 What is the value of the derivative of $y = 2 \sin 3x - 3 \tan x$ at $x = 0$?

- (A) -1
- (B) 0
- (C) 3
- (D) -9

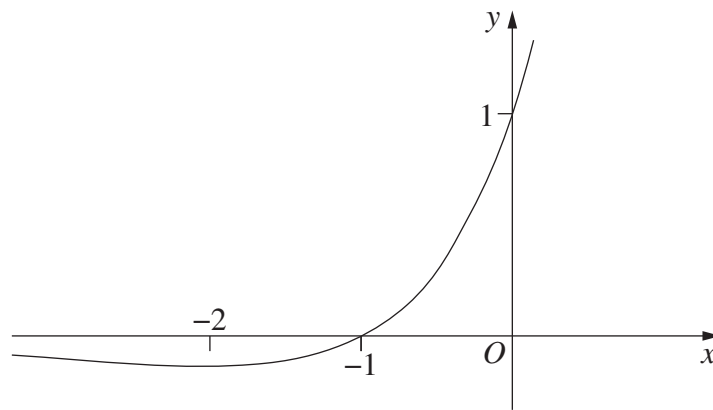
7 The diagram shows the parabola $y = 4x - x^2$ meeting the line $y = 2x$ at $(0, 0)$ and $(2, 4)$.



Which expression gives the area of the shaded region bounded by the parabola and the line?

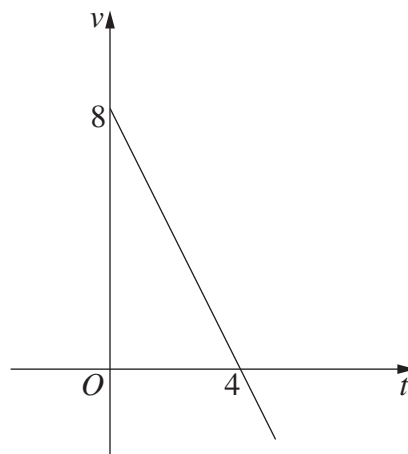
- (A) $\int_0^2 x^2 - 2x \, dx$
- (B) $\int_0^2 2x - x^2 \, dx$
- (C) $\int_0^4 x^2 - 2x \, dx$
- (D) $\int_0^4 2x - x^2 \, dx$

- 8 The diagram shows the graph of $y = e^x(1+x)$.



How many solutions are there to the equation $e^x(1+x) = 1-x^2$?

- (A) 0
(B) 1
(C) 2
(D) 3
- 9 A particle is moving along the x -axis. The graph shows its velocity v metres per second at time t seconds.

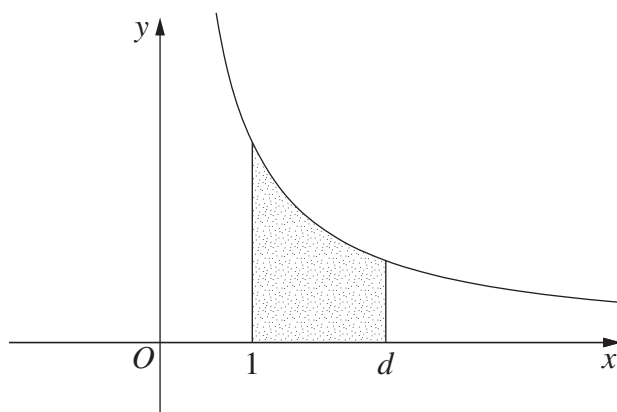


When $t = 0$ the displacement x is equal to 2 metres.

What is the maximum value of the displacement x ?

- (A) 8 m
(B) 14 m
(C) 16 m
(D) 18 m

- 10 The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$.



What value of d makes the shaded area equal to 2?

- (A) e
- (B) $e + 1$
- (C) $2e$
- (D) e^2

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

- (a) Simplify $4x - (8 - 6x)$. **1**
- (b) Factorise fully $3x^2 - 27$. **2**
- (c) Express $\frac{8}{2 + \sqrt{7}}$ with a rational denominator. **2**
- (d) Find the limiting sum of the geometric series $1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$. **2**
- (e) Differentiate $(e^x + x)^5$. **2**
- (f) Differentiate $y = (x + 4)\ln x$. **2**
- (g) Evaluate $\int_0^{\frac{\pi}{4}} \cos 2x \, dx$. **2**
- (h) Find $\int \frac{x}{x^2 - 3} \, dx$. **2**

Question 12 (15 marks) Use the Question 12 Writing Booklet.

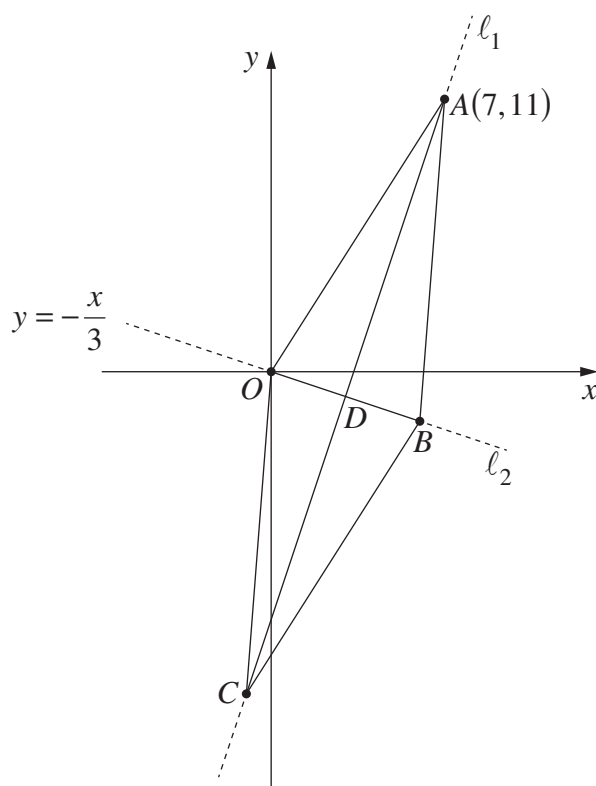
- (a) Find the solutions of $2 \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

2

- (b) The diagram shows the rhombus $OABC$.

The diagonal from the point $A(7, 11)$ to the point C lies on the line ℓ_1 .

The other diagonal, from the origin O to the point B , lies on the line ℓ_2 which has equation $y = -\frac{x}{3}$.



NOT TO
SCALE

- (i) Show that the equation of the line ℓ_1 is $y = 3x - 10$.

2

- (ii) The lines ℓ_1 and ℓ_2 intersect at the point D .

2

Find the coordinates of D .

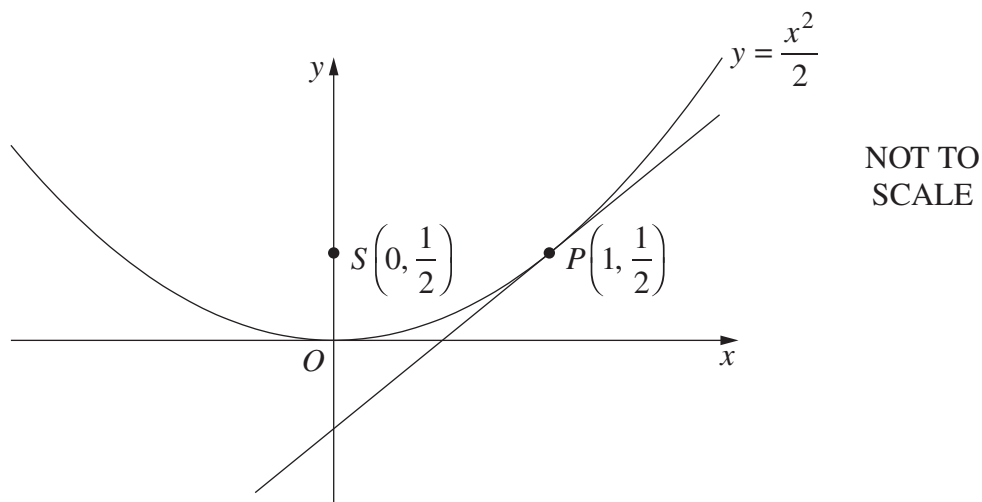
Question 12 continues on page 9

Question 12 (continued)

(c) Find $f'(x)$, where $f(x) = \frac{x^2 + 3}{x - 1}$. 2

(d) For what values of k does the quadratic equation $x^2 - 8x + k = 0$ have real roots? 2

(e) The diagram shows the parabola $y = \frac{x^2}{2}$ with focus $S\left(0, \frac{1}{2}\right)$. A tangent to the parabola is drawn at $P\left(1, \frac{1}{2}\right)$.

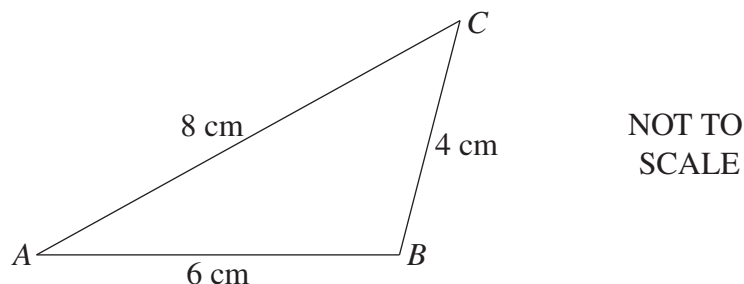


- (i) Find the equation of the tangent at the point P . 2
- (ii) What is the equation of the directrix of the parabola? 1
- (iii) The tangent and directrix intersect at Q .
 Show that Q lies on the y -axis. 1
- (iv) Show that $\triangle PQS$ is isosceles. 1

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

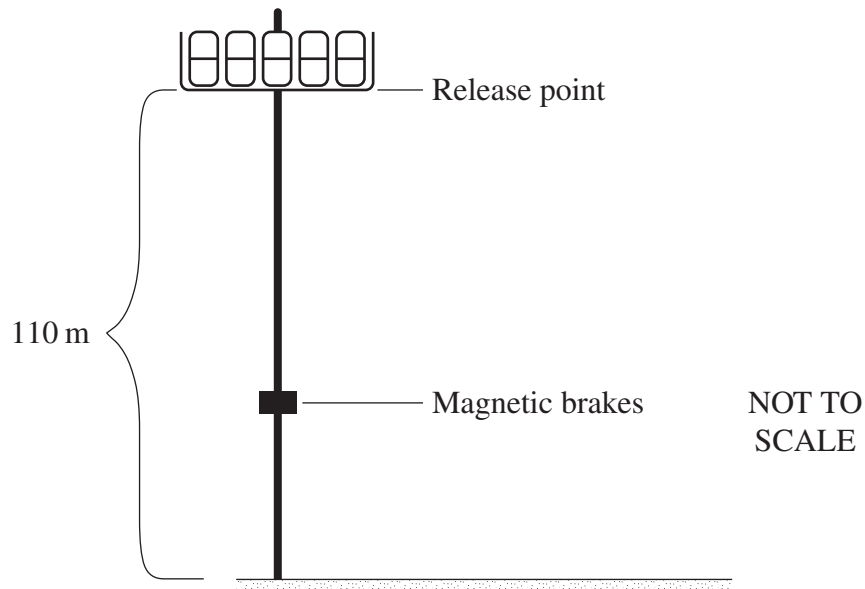
- (a) The diagram shows $\triangle ABC$ with sides $AB = 6$ cm, $BC = 4$ cm and $AC = 8$ cm.



- (i) Show that $\cos A = \frac{7}{8}$. **1**
- (ii) By finding the exact value of $\sin A$, determine the exact value of the area of $\triangle ABC$. **2**
- (b) (i) Find the domain and range for the function $f(x) = \sqrt{9 - x^2}$. **2**
- (ii) On a number plane, shade the region where the points (x, y) satisfy both of the inequalities $y \leq \sqrt{9 - x^2}$ and $y \geq x$. **2**
- (c) Consider the curve $y = x^3 - x^2 - x + 3$.
- (i) Find the stationary points and determine their nature. **4**
- (ii) Given that the point $P\left(\frac{1}{3}, \frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflexion at P . **2**
- (iii) Sketch the curve, labelling the stationary points, point of inflexion and y -intercept. **2**

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) In a theme park ride, a chair is released from a height of 110 metres and falls vertically. Magnetic brakes are applied when the velocity of the chair reaches -37 metres per second.



The height of the chair at time t seconds is x metres. The acceleration of the chair is given by $\ddot{x} = -10$. At the release point, $t = 0$, $x = 110$ and $\dot{x} = 0$.

- (i) Using calculus, show that $x = -5t^2 + 110$. **2**
- (ii) How far has the chair fallen when the magnetic brakes are applied? **2**

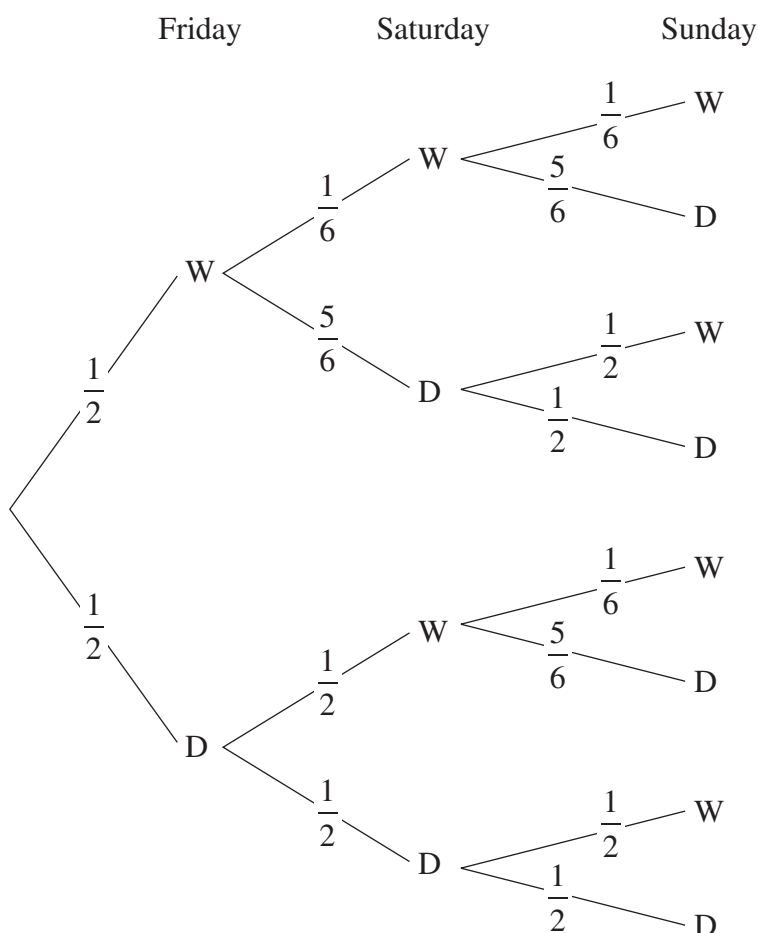
Question 14 continues on page 12

Question 14 (continued)

(b) Weather records for a town suggest that:

- if a particular day is wet (W), the probability of the next day being dry is $\frac{5}{6}$
- if a particular day is dry (D), the probability of the next day being dry is $\frac{1}{2}$.

In a specific week Thursday is dry. The tree diagram shows the possible outcomes for the next three days: Friday, Saturday and Sunday.



- (i) Show that the probability of Saturday being dry is $\frac{2}{3}$. 1
- (ii) What is the probability of both Saturday and Sunday being wet? 2
- (iii) What is the probability of at least one of Saturday and Sunday being dry? 1

Question 14 continues on page 13

Question 14 (continued)

- (c) Sam borrows \$100 000 to be repaid at a reducible interest rate of 0.6% per month. Let A_n be the amount owing at the end of n months and M be the monthly repayment.

(i) Show that $A_2 = 100\,000(1.006)^2 - M(1 + 1.006)$. **1**

(ii) Show that $A_n = 100\,000(1.006)^n - M\left(\frac{(1.006)^n - 1}{0.006}\right)$. **2**

- (iii) Sam makes monthly repayments of \$780. **1**

Show that after making 120 monthly repayments the amount owing is \$68 500 to the nearest \$100.

- (iv) Immediately after making the 120th repayment, Sam makes a one-off payment, reducing the amount owing to \$48 500. The interest rate and monthly repayment remain unchanged. **3**

After how many more months will the amount owing be completely repaid?

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) The amount of caffeine, C , in the human body decreases according to the equation

$$\frac{dC}{dt} = -0.14C,$$

where C is measured in mg and t is the time in hours.

- (i) Show that $C = Ae^{-0.14t}$ is a solution to $\frac{dC}{dt} = -0.14C$, where A is a constant. **1**

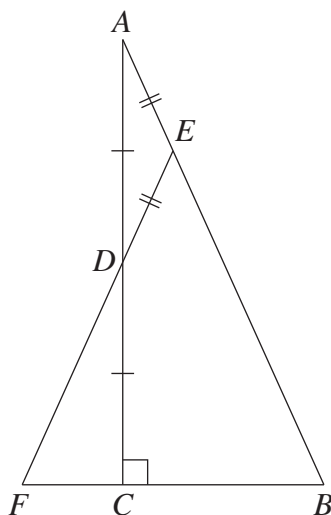
When $t = 0$, there are 130 mg of caffeine in Lee's body.

- (ii) Find the value of A . **1**
- (iii) What is the amount of caffeine in Lee's body after 7 hours? **1**
- (iv) What is the time taken for the amount of caffeine in Lee's body to halve? **2**

Question 15 continues on page 15

Question 15 (continued)

- (b) The diagram shows $\triangle ABC$ which has a right angle at C . The point D is the midpoint of the side AC . The point E is chosen on AB such that $AE = ED$. The line segment ED is produced to meet the line BC at F .



NOT TO
SCALE

Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) Prove that $\triangle ACB$ is similar to $\triangle DCF$. | 2 |
| (ii) Explain why $\triangle EFB$ is isosceles. | 1 |
| (iii) Show that $EB = 3AE$. | 2 |
- (c) Water is flowing in and out of a rock pool. The volume of water in the pool at time t hours is V litres. The rate of change of the volume is given by

$$\frac{dV}{dt} = 80\sin(0.5t).$$

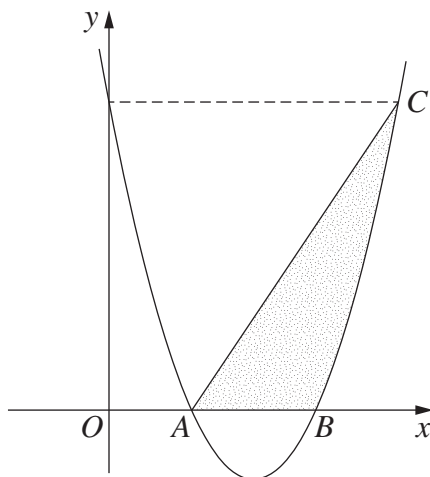
At time $t = 0$, the volume of water in the pool is 1200 litres and is increasing.

- | | |
|---|----------|
| (i) After what time does the volume of water first start to decrease? | 2 |
| (ii) Find the volume of water in the pool when $t = 3$. | 2 |
| (iii) What is the greatest volume of water in the pool? | 1 |

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) The diagram shows the curve with equation $y = x^2 - 7x + 10$. The curve intersects the x -axis at points A and B . The point C on the curve has the same y -coordinate as the y -intercept of the curve.

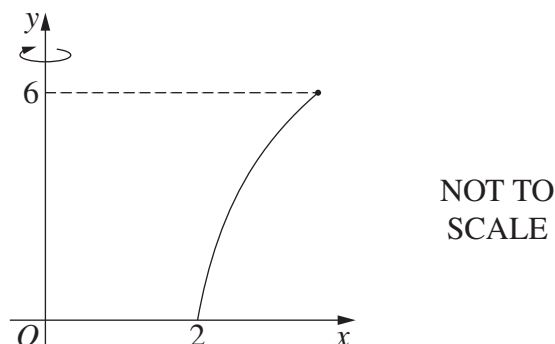


- | | | |
|-------|--|----------|
| (i) | Find the x -coordinates of points A and B . | 1 |
| (ii) | Write down the coordinates of C . | 1 |
| (iii) | Evaluate $\int_0^2 (x^2 - 7x + 10) dx$. | 1 |
| (iv) | Hence, or otherwise, find the area of the shaded region. | 2 |

Question 16 continues on page 17

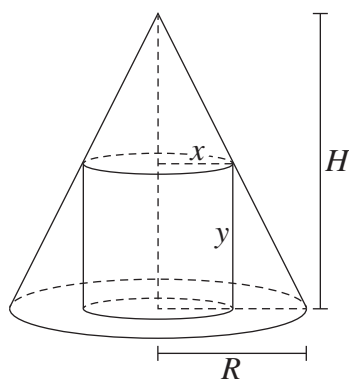
Question 16 (continued)

- (b) A bowl is formed by rotating the curve $y = 8 \log_e(x-1)$ about the y -axis for $0 \leq y \leq 6$. **3**



Find the volume of the bowl. Give your answer correct to 1 decimal place.

- (c) The diagram shows a cylinder of radius x and height y inscribed in a cone of radius R and height H , where R and H are constants.



The volume of a cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.

The volume of a cylinder of radius r and height h is $\pi r^2 h$.

- (i) Show that the volume, V , of the cylinder can be written as **3**
- $$V = \frac{H}{R} \pi x^2 (R - x).$$

- (ii) By considering the inscribed cylinder of maximum volume, show that the volume of any inscribed cylinder does not exceed $\frac{4}{9}$ of the volume of the cone. **4**

End of paper

BLANK PAGE

BLANK PAGE

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2015 HSC Mathematics Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	C
3	A
4	A
5	B
6	C
7	B
8	C
9	D
10	A

Section II

Question 11 (a)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 &4x - (8 - 6x) \\
 &= 4x - 8 + 6x \\
 &= 10x - 8
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Finds common factor of 3, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 &3x^2 - 27 \\
 &= 3(x^2 - 9) \\
 &= 3(x - 3)(x + 3)
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $2 - \sqrt{7}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{8}{2 + \sqrt{7}} &= \frac{8}{2 + \sqrt{7}} \times \frac{2 - \sqrt{7}}{2 - \sqrt{7}} \\
 &= \frac{8(2 - \sqrt{7})}{4 - 7} \\
 &= \frac{8(2 - \sqrt{7})}{-3} \\
 &= \frac{8\sqrt{7} - 16}{3}
 \end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Identifies the common ratio, or equivalent merit	1

Sample answer:

$$a = 1 \quad r = -\frac{1}{4}$$

$$\begin{aligned}
 S &= \frac{a}{1-r} \\
 &= \frac{1}{1 - \left(-\frac{1}{4}\right)} \\
 &= \frac{4}{5}
 \end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use chain rule, or equivalent merit	1

Sample answer:

$$\text{Let } y = (e^x + x)^5$$

$$\frac{dy}{dx} = 5(e^x + x)^4 \times (e^x + 1)$$

$$= 5(e^x + 1)(e^x + x)^4$$

Question 11 (f)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use product rule, or equivalent merit	1

Sample answer:

$$\text{Let } u = x + 4 \quad v = \ln x$$

$$u' = 1 \quad v' = \frac{1}{x}$$

$$\therefore y' = u'v + v'u$$

$$= 1 \cdot \ln x + \frac{1}{x}(x + 4)$$

$$= \ln x + \frac{x + 4}{x}$$

Question 11 (g)

Criteria	Marks
• Provides correct solution	2
• Provides correct primitive, or equivalent merit	1

Sample answer:

$$\int_0^{\frac{\pi}{4}} \cos 2x \, dx = \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}$$

Question 11 (h)

Criteria	Marks
• Provides correct primitive	2
• Recognises $\frac{f'(x)}{f(x)}$, or equivalent merit	1

Sample answer:

$$\begin{aligned} & \int \frac{x}{x^2 - 3} dx \\ &= \frac{1}{2} \int \frac{2x}{x^2 - 3} dx \\ &= \frac{1}{2} \ln(x^2 - 3) + C \end{aligned}$$

Question 12 (a)

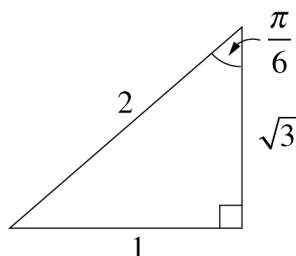
Criteria	Marks
• Provides correct solutions	2
• Provides one correct solution, or equivalent merit	1

Sample answer:

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

**Question 12 (b) (i)**

Criteria	Marks
• Provides correct solution	2
• Attempts to use the fact that the diagonals are perpendicular, or equivalent merit	1

Sample answer:

$$\ell_1 \text{ is } y = m_1 x + b$$

$$\ell_2 \text{ is } y = -\frac{x}{3} \Rightarrow m_2 = -\frac{1}{3}$$

$$\ell_1 \perp \ell_2 \quad OABC \text{ rhombus (diagonals } \ell_1, \ell_2 \text{ are perpendicular)}$$

$$m_1 m_2 = -1$$

$$m_1 \times \left(-\frac{1}{3}\right) = -1$$

$$\therefore m_1 = 3$$

$$\ell_1 \text{ is } y = 3x + b$$

$$11 = 3 \times 7 + b$$

$$b = -10$$

$$\therefore y = 3x - 10$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct coordinates	2
• Makes some progress towards correct answer	1

Sample answer:

Solving $y = -\frac{x}{3}$

and $y = 3x - 10$ simultaneously

$$-\frac{x}{3} = 3x - 10$$

$$x = -9x + 30$$

$$10x = 30$$

$$x = 3$$

Hence, $y = -\frac{3}{3}$

$$= -1$$

$$\therefore D(3, -1)$$

Question 12 (c)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use the quotient rule, or equivalent merit	1

Sample answer:

Let $u = x^2 + 3$ $v = x - 1$

$$u' = 2x \quad v' = 1$$

$$y' = \frac{u'v - v'u}{v^2}$$

$$= \frac{2x(x-1) - 1(x^2 + 3)}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2 - 3}{(x-1)^2}$$

$$= \frac{x^2 - 2x - 3}{(x-1)^2}$$

Question 12 (d)

Criteria	Marks
• Provides correct solution	2
• Evaluates discriminant, or equivalent merit	1

Sample answer:

For real roots, $\Delta \geq 0$

$$\Delta = b^2 - 4ac$$

$$= (-8)^2 - 4(1)(k)$$

$$= 64 - 4k$$

$$\therefore 64 - 4k \geq 0$$

$$64 \geq 4k$$

$$k \leq 16$$

Question 12 (e) (i)

Criteria	Marks
• Provides a correct equation	2
• Finds correct gradient, or equivalent merit	1

Sample answer:

$$y = \frac{x^2}{2} \quad P\left(1, \frac{1}{2}\right)$$

$$\frac{dy}{dx} = x$$

$$\text{when } x = 1, \quad m = 1$$

The equation of the tangent is

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{2} = 1(x - 1)$$

$$\therefore y = x - \frac{1}{2}$$

Question 12 (e) (ii)

Criteria	Marks
• Provides a correct equation	1

Sample answer:

$$\text{Focus } S(0, a) = \left(0, \frac{1}{2}\right) \therefore a = \frac{1}{2}$$

$$\text{Equation directrix is: } y = -a$$

$$\therefore y = -\frac{1}{2}$$

Question 12 (e) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\text{Tangent: } y = x - \frac{1}{2}$$

$$\text{Directrix: } y = -\frac{1}{2}$$

$$\begin{aligned} \text{Solution: } -\frac{1}{2} &= x - \frac{1}{2} \\ x &= 0 \end{aligned}$$

$\therefore Q$ lies on y-axis

Question 12 (e) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$SQ = SO + OQ$$

$$= \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

$$PS = 1$$

$$\therefore SQ = PS$$

$\therefore \triangle PQS$ is isosceles (two equal sides)

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{8^2 + 6^2 - 4^2}{2 \times 8 \times 6}$$

$$= \frac{84}{96}$$

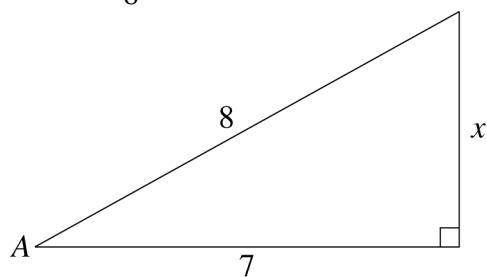
$$= \frac{7}{8}$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds exact value of $\sin A$, or equivalent merit	1

Sample answer:

$$\cos A = \frac{7}{8}$$



$$x^2 = \sqrt{8^2 - 7^2}$$

$$x = \sqrt{15}$$

$$\therefore \sin A = \frac{\sqrt{15}}{8}$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2}bc \sin A$$

$$= \frac{1}{2} \times 8 \times 6 \times \frac{\sqrt{15}}{8}$$

$$\therefore \text{Area} = 3\sqrt{15} \text{ cm}^2$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Identifies -3 and 3 as important values, or equivalent merit	1

Sample answer:

Domain: $9 - x^2 \geq 0$

$$x^2 \leq 9$$

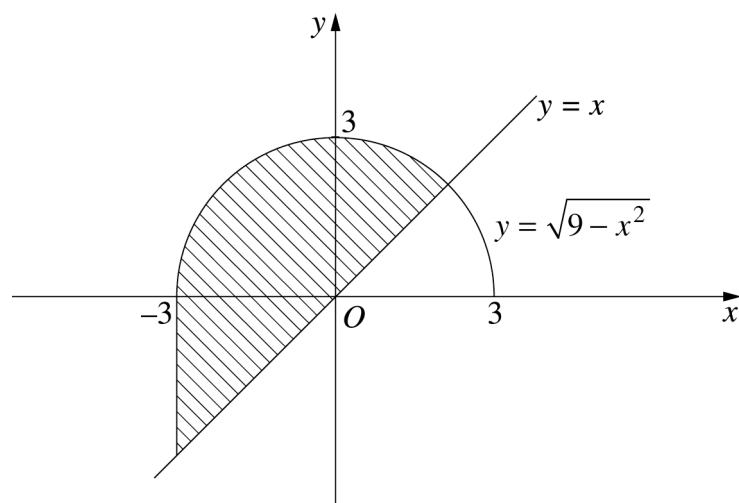
$$-3 \leq x \leq 3$$

Range: $0 \leq y \leq 3$

Question 13 (b) (ii)

Criteria	Marks
• Shades correct region	2
• Shades a region involving the correct semicircle, or equivalent merit	1

Sample answer:



Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	4
• Finds coordinates of both stationary points, or equivalent merit	3
• Factorises quadratic in $\frac{dy}{dx} = 0$ correctly, or equivalent merit	2
• Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit	1

Sample answer:

$$y = x^3 - x^2 - x + 3$$

$$y' = 3x^2 - 2x - 1$$

$$y'' = 6x - 2$$

Stationary points when $y' = 0$ ie $3x^2 - 2x - 1 = 0$

$$(3x + 1)(x - 1) = 0$$

$$\therefore x = -\frac{1}{3} \text{ or } x = 1$$

when $x = -\frac{1}{3}$, $y = 3\frac{5}{27}$ and $y'' = -4 < 0$ (cd)

$\therefore \left(-\frac{1}{3}, 3\frac{5}{27}\right)$ is a local maximum

when $x = 1$, $y = 2$ and $y'' = 4 > 0$ (cu)

$\therefore (1, 2)$ is a local minimum

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows second derivative = 0 at $x = \frac{1}{3}$, or equivalent merit	1

Sample answer:

$$P\left(\frac{1}{3}, \frac{70}{27}\right)$$

Point of inflexion where $y'' = 0$ and concavity changes

$$\text{ie } 6x - 2 = 0$$

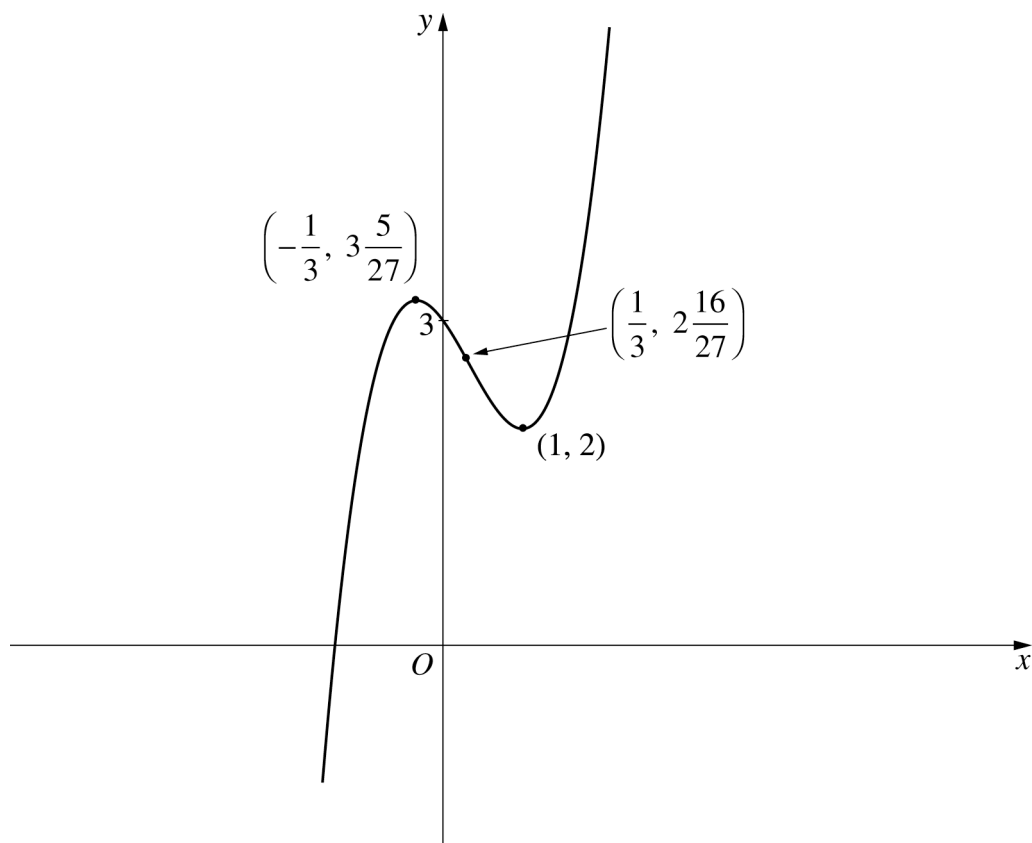
$$x = \frac{1}{3}$$

There is a point of inflexion at $x = \frac{1}{3}$ since from (i) concavity changes (maximum at $x = -\frac{1}{3}$ and minimum at $x = 1$).

Question 13 (c) (iii)

Criteria	Marks
• Provides correct sketch	2
• Correctly plots and labels stationary points and point of inflexion, or equivalent merit	1

Sample answer:



Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct expression for $\dot{x}$ including determination of the constant, or equivalent merit	1

Sample answer:

$$\ddot{x} = -10, \quad t = 0, \quad x = 110, \quad \dot{x} = 0$$

$$\dot{x} = \int -10 \, dt$$

$$= -10t + C_1$$

When $t = 0$, $\dot{x} = 0$

$$\therefore 0 = -10(0) + C_1$$

$$\therefore C_1 = 0$$

$$\therefore \dot{x} = -10t$$

$$\text{Now, } x = \int -10t \, dt$$

$$= -5t^2 + C_2$$

When $t = 0$, $x = 110$

$$\therefore 110 = -5(0)^2 + C_2$$

$$\therefore C_2 = 110$$

$$\therefore x = -5t^2 + 110$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds correct time when brakes are applied, or equivalent merit	1

Sample answer:When $\dot{x} = -37$

$$-37 = -10t$$

$$t = \frac{37}{10}$$

$$\begin{aligned}\therefore x &= -5\left(\frac{37}{10}\right)^2 + 110 \\ &= 41.55\end{aligned}$$

$$\begin{aligned}\therefore \text{It has fallen } (110 - 41.55) \text{ m} \\ = 68.45 \text{ m}\end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$P(\text{Saturday dry}) = P(WD) + P(DD)$$

$$\begin{aligned}&= \frac{1}{2} \times \frac{5}{6} + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{2}{3}\end{aligned}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Provides correct expression for probability of one of Saturday or Sunday being wet, or equivalent merit	1

Sample answer: $P(\text{both Saturday and Sunday wet})$

$$= P(WWW) + P(DWW)$$

$$= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{72} + \frac{1}{24}$$

$$= \frac{1}{18}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer: $P(\text{at least one of Saturday and Sunday dry})$

$$= 1 - P(\text{both are wet})$$

$$= 1 - \frac{1}{18}$$

$$= \frac{17}{18}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$r = 0.6\% \text{ per month}$$

$$= 0.006 \text{ per month}$$

$$A_1 = 100\,000(1.006) - M$$

$$A_2 = A_1(1.006) - M$$

$$= (100\,000(1.006) - M)(1.006) - M$$

$$A_2 = 100\,000(1.006)^2 - M(1.006) - M$$

$$= 100\,000(1.006)^2 - M(1 + 1.006)$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Recognises the pattern in the expression for A_n as a sum, or equivalent merit	1

Sample answer:

$$A_3 = A_2(1.006) - M$$

$$= 100\,000(1.006)^3 - M(1.006)^2 - M(1.006) - M$$

$$\vdots$$

$$A_n = 100\,000(1.006)^n - M[1 + (1.006) + (1.006)^2 + \dots + (1.006)^{n-1}]$$

Now $[1 + (1.006) + (1.006)^2 + \dots + (1.006)^{n-1}]$ isa geometric series, with n terms where $a = 1$ and $r = 1.006$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1((1.006)^n - 1)}{(1.006) - 1}$$

$$\therefore A_n = 100\,000(1.006)^n - M \left[\frac{1((1.006)^n - 1)}{0.006} \right]$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$M = 780, n = 120$$

$$A_{120} = 100\,000(1.006)^{120} - 780 \left[\frac{(1.006)^{120} - 1}{0.006} \right]$$

$$= 68\,499.46$$

$\therefore$ amount owing = \$68 500 (nearest \$100)

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	3
• Finds correct expression for $(1.006)^n$, or equivalent merit	2
• Attempts to use $A_n = 0$, or equivalent merit	1

Sample answer:

Since the amount owing is \$48 500,

$$A_n = 48\,500(1.006)^n - 780 \left[\frac{(1.006)^n - 1}{0.006} \right]$$

Let $A_n = 0$ when the amount owing is completely repaid

$$\therefore 48\,500(1.006)^n = 780 \left[\frac{(1.006)^n - 1}{0.006} \right]$$

$$\therefore 291(1.006)^n = 780(1.006)^n - 780$$

$$\therefore 489(1.006)^n = 780$$

$$\therefore (1.006)^n = 1.59509\dots$$

$$n \log(1.006) = \log(1.59509\dots)$$

$$n \approx \frac{\log(1.59509)}{\log 1.006}$$

time required ≈ 78.055 months

Question 15 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$C = Ae^{-0.14t}$$

$$\begin{aligned}\frac{dC}{dt} &= -0.14Ae^{-0.14t} \\ &= -0.14C\end{aligned}$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct value of A	1

Sample answer:When $t = 0$, $C = 130$

$$130 = Ae^0$$

$$\therefore A = 130$$

Question 15 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$C = 130e^{-0.14t}$$

When $t = 7$,

$$C = 130e^{-0.14 \times 7}$$

$$\approx 48.79044285$$

amount of caffeine ≈ 49 mg

Question 15 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to use logarithms to solve $65 = 130e^{-0.14t}$, or equivalent merit	1

Sample answer:Taking $C = 65$

$$65 = 130e^{-0.14t}$$

$$\frac{1}{2} = e^{-0.14t}$$

$$\ln \frac{1}{2} = -0.14t$$

$$t = \frac{\ln \frac{1}{2}}{-0.14}$$

$$= 4.95105129$$

time taken ≈ 4.95 hours

Question 15 (b) (i)

Criteria	Marks
• Provides correct proof	2
• Shows $\angle FDC$ equal to $\angle ADE$, or $\angle ADE$ equal to $\angle DAE$, or equivalent merit	1

Sample answer:

$$\angle FDC = \angle ADE \quad (\text{vertically opposite angles equal})$$

$$\angle ADE = \angle DAE \quad (\text{base angles, isosceles triangle})$$

$$\therefore \angle FDC = \angle DAE$$

$$\begin{aligned} \angle FCD &= \angle ACB \\ &= 90^\circ \quad (\text{given}) \end{aligned}$$

$$\therefore \triangle ACB \parallel \triangle DCF \quad (\text{equal angles})$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle DFC = \angle ABC \quad (\text{corresponding angles in similar triangles})$$

$$\therefore \triangle EFB \text{ is isosceles} \quad (\text{base angles equal})$$

Question 15 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Provides correct relationship between lengths of AB and FD, or drops perpendicular from E onto AD, or equivalent merit	1

Sample answer:

$$AB = 2FD \quad (\text{using (i), corresponding sides in proportion } AC = 2DC)$$

$$\text{ie } AE + EB = 2FD$$

$$\text{Also, } FD = EF - ED$$

$$\text{so } FD = EB - AE \quad (\text{by (ii) since } EF = EB)$$

$$\text{Hence } AE + EB = 2(EB - AE)$$

$$\therefore EB = 3AE$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $\frac{dV}{dt} = 0$, or equivalent merit	1

Sample answer:

Set $\frac{dV}{dt} = 0$

$$80\sin(0.5t) = 0$$

$$\sin 0.5t = 0$$

$$0.5t = 0, \pi, 2\pi, \dots$$

first starts to decrease when $0.5t = \pi$

$$\text{so } t = 2\pi$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Provides correct primitive, or equivalent merit	1

Sample answer:

$$\frac{dV}{dt} = 80\sin(0.5t)$$

Integrating,

$$V = -160\cos(0.5t) + C$$

When $t = 0$, $1200 = -160\cos(0.5 \times 0) + C$

$$C = 1360$$

Hence, $V = -160\cos(0.5t) + 1360$

When $t = 3$, $V = -160\cos(0.5 \times 3) + 1360$

Volume ≈ 1349 L

Question 15 (c) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Greatest volume when $\frac{dV}{dt} = 0$

ie when $t = 2\pi$ and volume starts to decrease

$$\begin{aligned} V &= -160\cos(0.5 \times 2\pi) + 1360 \\ &= 1520 \end{aligned}$$

volume = 1520 L

Question 16 (a) (i)

Criteria	Marks
• Provides correct x -coordinates	1

Sample answer:

$$\text{When } y = 0, x^2 - 7x + 10 = 0$$

$$\Rightarrow (x - 2)(x - 5) = 0$$

$$\therefore x = 2, x = 5$$

So A is $(2,0)$ and B is $(5,0)$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct coordinates	1

Sample answer:

$$\text{When } x = 0, y = 10$$

$$\Rightarrow C \text{ is } (7,10) \quad (\text{by symmetry})$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\int_0^2 (x^2 - 7x + 10) dx = \left[\frac{x^3}{3} - \frac{7x^2}{2} + 10x \right]_0^2$$

$$= \left(\frac{2^3}{3} - 7 \times \frac{2^2}{2} + 10 \times 2 \right) - \left(\frac{0^3}{3} - 7 \times \frac{0^2}{2} + 10 \times 0 \right)$$

$$= \frac{26}{3}$$

Question 16 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to find the difference of two areas or integrals, or equivalent merit	1

Sample answer:

By symmetry, $\int_0^2 (x^2 - 7x + 10) dx = \int_5^7 (x^2 - 7x + 10) dx$

$\therefore$ Shaded area = area of $\triangle - \int_5^7 (x^2 - 7x + 10) dx$

$$= \frac{1}{2} \times (7 - 2) \times 10 - \frac{26}{3}$$

$$= \frac{49}{3} \text{ units}^2$$

Question 16 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains $\pi \int_0^6 \left(e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1 \right) dy$, or equivalent merit	2
• Writes down formula for the volume including limits OR • Writes x as a function of y , or equivalent merit	1

Sample answer:

$$V = \pi \int_0^6 x^2 dy$$

$$\text{Now } y = 8 \log_e (x - 1)$$

$$\Rightarrow e^{\frac{y}{8}} = x - 1$$

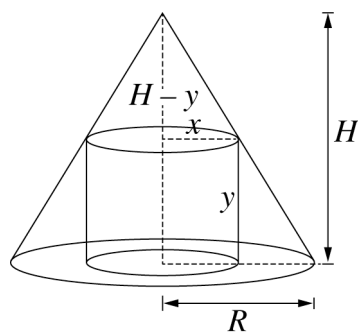
$$\text{ie } x = e^{\frac{y}{8}} + 1$$

$$\begin{aligned} \therefore V &= \pi \int_0^6 \left(e^{\frac{y}{8}} + 1 \right)^2 dy \\ &= \pi \int_0^6 \left(e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1 \right) dy \\ &= \pi \left[4e^{\frac{y}{4}} + 16e^{\frac{y}{8}} + y \right]_0^6 \\ &= \pi \left[\left(4e^{\frac{3}{2}} + 16e^{\frac{3}{4}} + 6 \right) - (4 + 16 + 0) \right] \\ &= \pi \left[4e^{\frac{3}{2}} + 16e^{\frac{3}{4}} - 14 \right] \end{aligned}$$

$$\text{volume} \approx 118.7$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Uses similar triangles to obtain correct equation relating x and y , or equivalent merit	2
• Attempts to use similar triangles, or equivalent merit	1

Sample answer:

Using similar triangles

$$\frac{H-y}{H} = \frac{x}{R} \quad \left[\text{OR, } \frac{H}{R} = \frac{y}{R-x} \right]$$

$$RH - Ry = xH$$

$$RH - xH = Ry$$

$$y = \frac{H(R-x)}{R}$$

$$\therefore V = \pi r^2 h = \pi x^2 y$$

$$= \pi x^2 \frac{H}{R} (R-x)$$

$$= \frac{H}{R} \pi x^2 (R-x)$$

Question 16 (c) (ii)

Criteria	Marks
• Provides correct solution	4
• Finds the value at which the stationary point occurs and verifies it is a maximum, or equivalent merit	3
• Finds the value of x at which $\frac{dV}{dx} = 0$, or equivalent merit	2
• Finds correct expression for $\frac{dV}{dx}$, or equivalent merit	1

Sample answer:

Maximum V occurs when $\frac{dV}{dx} = 0$ and $\frac{d^2V}{dx^2} < 0$

$$V = \frac{H}{R}\pi x^2(R-x) = H\pi x^2 - \frac{H}{R}\pi x^3$$

$$\frac{dV}{dx} = 2H\pi x - \frac{3H}{R}\pi x^2 = 0 \text{ for maximum}$$

$$\therefore 0 = \pi Hx \left(2 - \frac{3}{R}x \right)$$

$$x = 0 \text{ (discount)} \text{ or } x = \frac{2R}{3}$$

$$\frac{d^2V}{dx^2} = 2H\pi - \frac{6H}{R}\pi x$$

$$= 2H\pi - \frac{6H}{R}\pi \frac{2R}{3} \text{ at } x = \frac{2R}{3}$$

$$= 2H\pi - 4H\pi$$

$$= -2H\pi < 0 \text{ since } H > 0$$

$$\therefore \text{Maximum volume when } x = \frac{2R}{3}$$

$$\therefore V = \frac{H}{R}\pi \left(\frac{2R}{3} \right)^2 \left(R - \frac{2R}{3} \right)$$

$$= \frac{H}{R}\pi \frac{4R^2}{9} \left(\frac{R}{3} \right)$$

$$V = \frac{4H\pi R^2}{27}$$

2015 HSC Mathematics

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.1	P3, P4
2	1	6.2	P4
3	1	7.1	H5
4	1	3.3	H5
5	1	11.3	H4
6	1	13.4, 13.5, 13.7	H5
7	1	11.4	H8
8	1	9.1, 12.1	H3
9	1	14.3	H4, H5
10	1	12.5	H3, H8

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	1	1.3	P3
11 (b)	2	1.3	P4
11 (c)	2	1.1	P3
11 (d)	2	7.3	H5
11 (e)	2	8.9, 12.5	P7, H3
11 (f)	2	8.9, 12.5	P7, H3
11 (g)	2	11.1, 13.7	H5
11(h)	2	11.2, 12.5	H5
12 (a)	2	5.3, 13.1	P4, H5
12 (b) (i)	2	2.3, 6.8	H5
12 (b) (ii)	2	1.4, 6.3, 6.8	P4, H5
12 (c)	2	8.8	P7
12 (d)	2	1.4, 9.2, 9.3	P4
12 (e) (i)	2	6.2, 8.5	P4
12 (e) (ii)	1	9.5	P4, P5
12 (e) (iii)	1	1.4, 6.3	P4, P5
12 (e) (iv)	1	2.2, 6.8	P4, H2, H5
13 (a) (i)	1	5.5	P3, P4
13 (a) (ii)	2	5.5	P3, P4

Question	Marks	Content	Syllabus outcomes
13 (b) (i)	2	4.1	P5
13 (b) (ii)	2	4.1, 4.4	P5
13 (c) (i)	4	10.2, 10.4	H6
13 (c) (ii)	2	10.4	H6
13 (c) (iii)	2	10.5	H6
14 (a) (i)	2	14.3	H4, H5
14 (a) (ii)	2	14.3	H4, H5
14 (b) (i)	1	3.2, 3.3	H5
14 (b) (ii)	2	3.2, 3.3	H5
14 (b) (iii)	1	3.2, 3.3	H5
14 (c) (i)	1	7.5	H4, H5
14 (c) (ii)	2	7.2, 7.5	H4, H5
14 (c) (iii)	1	7.5	H4, H5
14 (c) (iv)	3	7.5, 12.2	H3, H5
15 (a) (i)	1	14.2	H3, H5
15 (a) (ii)	1	14.2	H3
15 (a) (iii)	1	14.2	H3
15 (a) (iv)	2	12.2, 14.2	H3
15 (b) (i)	2	2.3, 2.5	H2, H5
15 (b) (ii)	1	2.2, 2.5	H2, H5
15 (b) (iii)	2	2.3, 2.5	H2, H5
15 (c) (i)	2	14.1	H4, H5
15 (c) (ii)	2	14.1	H4, H5
15 (c) (iii)	1	14.1	H4, H5
16 (a) (i)	1	9.1	P4
16 (a) (ii)	1	1.4, 4.2	P4
16 (a) (iii)	1	11.1	H8
16 (a) (iv)	2	11.4	H8, H9
16 (b)	3	11.4, 12.3, 12.5	H3, H4, H8
16 (c) (i)	3	2.3, 2.5	P4, P5
16 (c) (ii)	4	10.6	H5